

KEY QUALIFICATIONS

3+ years in ML and software development with a record of improving system efficiency; proficient in Python, Java, and SQL with expertise in secure ETL, data integration, automated ML workflows, data visualization, and cloud platforms. MS-level research applying probabilistic neural networks to emerging device modeling.

SKILLS SUMMARY

- Python/Java/C
- SQL
- Tableau/Power BI
- AWS/Azure
- TensorFlow

PROFESSIONAL EXPERIENCE & PROJECTS

Graduate Research Assistant

University of Tennessee Knoxville, NorDIC Lab

Aug 2023 – Present

Knoxville, TN, United States

Built ML-driven probabilistic, circuit-simulator-compatible models to capture statistical variation in semiconductor devices, improved prediction accuracy by **20%** compared to baseline models. Developed a predictive ML model for inverse design that maps desired outputs to optimal input parameters.

Data Analyst

Naztech Inc. Ltd.

Jan 2019 – Jan 2022

Dhaka, Bangladesh

- Built and optimized **ETL pipelines** that processed over **1M** records weekly, integrating into a **cloud** environment, including automated data cleaning and review, and connecting to **Power BI** and **Tableau** dashboards for business reporting.
- Reduced manual reporting effort by **70%**, improved data accuracy to **80%**, and delivered interactive insights for business users in real time.

Software Engineering Intern

Robi Axiata Ltd.

Jun 2017 – Sep 2017

Dhaka, Bangladesh

- Resolved application bugs using **Python**, **Git**, **SQLite** & **AWS CloudWatch**, leading to a **15%** reduction in customer complaints and improved stability.
- Performed QA/QC review of application data and logs to ensure accuracy, consistency, and reliability of bug fixes and system updates.

Machine Learning-Based Compact Model: Building a compact model using LSTM layers to integrate phase-field modeling with SPICE simulations, enhancing prediction accuracy for nanoelectronic devices.

Crop Prediction Using Machine Learning and Data Analysis: Conducted extensive **data analysis** on 250K+ agricultural records (12 features), performing **data cleaning, feature engineering, scaling, and dimensionality reduction**. Built predictive models based on geo/climate using Python, and TensorFlow/Keras, achieving up to **95%** accuracy with **ANN**, **SVM**, and **k-NN**.

A portable single-cell Braille display: Developed using **Arduino (C/C++)** with **embedded programming** to parse serial data, map characters, and control actuators via state machines. Achieved **97%** accuracy and reduced per-cell cost to \$29 (≈30% cheaper)

CONFERENCES

- IEEE Computer Society Annual Symposium on VLSI (ISVLSI) (Tennessee, USA, 2024):** Machine Learning-Based Compact Modeling Framework for Nanoelectronic Devices.

PROFESSIONAL TRAINING

IT Training, Certified by George Washington University

Conducted by Ernst & Young, India under LICT Project of BD Computer Council

2018 (Permanent)

Dhaka, Bangladesh

EDUCATION

University of Tennessee Knoxville

MS - Computer Engineering; CGPA: 3.96/4.00

TN, United States

Jul 2025

BRAC University

BS (Hons.) - Computer Science

Dhaka, Bangladesh

Dec 2018